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Ground subsidence observations and a modified influence function method for complete subsidence prediction

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Abstract

The results of surface subsidence investigations carried out in different coal fields in India are presented in this paper. Important differences are seen to exist between the subsidence norms given in the Subsidence Engineers Handbook, UK and those obtained in Indian conditions. Differences also exist for extraction below unfractured or uncaved rock and below pre-existing caved goaf(s). The subsidence is higher in the latter case and also in the slopes and strains. Consequently separate equations and guidelines have been given for these two conditions of mining.

Since Indian coal mining is mostly under shallow to moderate cover, several cases of discontinuous subsidence have been observed. A demarcation limit for discontinuous subsidence is given, seeing that discontinuous subsidence causes severe surface damage.

Since room and pillar mining is the principal underground method of extraction, panels of irregular shapes often occur making complete subsidence prediction difficult. The influence function method with a new function and additional functions for modelling subsidence asymmetry and the effect of extraction edges are given in this paper. The method for extraction below goaf(s) is also examined. Four cases are given for the comparison of observed and predicted subsidence contours and two cases for comparison of the slopes and strains. © 2000 Elsevier Science Ltd. All rights reserved.

1. Introduction

Mine subsidence studies at the Central Mining Research Institute (CMRI), Dhanbad have been going on for the last 35 years in different coal fields of India. The principal method of underground coal extraction is room and pillar mining, although longwall and some special methods of thick seam extraction have also been practised to some extent. Room and pillar mining often gives rise to extraction panels of irregular shapes making subsidence predictions difficult because all earlier empirical approaches have been two-dimensional. In fact, such terms as width-to-depth ratio of extraction, critical width of extraction, subsidence profile and even angle of draw tacitly assume a plane strain condition.

Of all the different tools available for subsidence modelling, the influence function method with suitable modifications appears to be a powerful method for complete subsidence prediction for all shapes of extraction panels. This method, in its classical form, consists of laying a surface grid of square elements overlapping the panel up to the draw limits. An influence circle, subdivided into a sufficient number of rings and sectors, gives the subsidence at the grid point which is the sum of the weighting factors of all the sectors falling within the extracted area. The weighting factors are derived from an axisymmetric influence function of bell shape covering the influence circle. The classical method, however, requires some modifications to suit the observed subsidence behaviour. Given in the following pages, are the salient findings from subsidence observations in

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Nomenclature

a	subsidence factor
A	area of influence annulus
A_0	area of influence circle
d	distance of extraction element centroid from starting face line
d_b	distance of extraction element to nearest boundary
d_{max}	distance between starting and ending face line
e	extraction (expressed as percentage or fraction)
$E_{(-)}$	maximum compressive strain
$E_{(+)}$	maximum tensile strain
G	maximum slope
H	depth of cover
K_1, K_2, K_3	constants in equations for maximum slope, compressive strain and tensile strain, respectively
K_z	influence function for subsidence
L	width of panel
NEW	non-effective width-to-depth ratio
r	radial distance of sector centroid from center of influence circle
R	radius of influence circle
S	maximum subsidence
S_i	weighting factor for ith ring
S_{ie}	weighting factor for each element in the ith ring
S_{max}	maximum possible subsidence (when the panel width is critical or super critical)
T	parting thickness between seams
V	displacement

Greek symbols

α	angle subtended at grid point P between x axis and centroid of each sector angle subtended at grid point P between x axis
β	centroid of extraction element
θ	angle of draw
ζ	angle between vertical and line joining element centroid with P

India, followed by a modified influence function method. These subsidence studies were carried out largely over room and pillar panels, although some longwall panels were also covered.

Numerical techniques offer some possibilities for subsidence modelling, but these are invariably fraught with some difficulties. Continuum methods, (finite or boundary element, fast Lagrangian analysis of continua, displacement discontinuity) fail completely, even with elasto-plastic analysis performed using realistic rock mass strength inputs, assuming that such inputs are possible for each bed. The distinct element method will possibly be the best, but again, this will require rock characterisation of each bed and this is usually not possible. One aspect, which is generally ignored in numerical modelling, is the possibility of each bed having different horizontal in situ stresses [1]. How this will affect rock failure and subsequent subsidence is a very dark area indeed. Because of these difficulties, the influence function method has been resorted to as a more convenient and simpler alternative.

With some exceptions, underground coal mining in India is confined to shallow or moderate depths (less than 350 m). This has given rise to several cases of discontinuous or stepped subsidence. Another important point is that because of manual, or at most semi-mechanised, mining room and pillar panels, tend to be small (commensurate with the incubation period), so that it is often difficult to get critical or super-critical widths of extraction. Such widths can exist under shallow covers but then discontinuous subsidence over such panels takes place.

Subsidence norms given in the Subsidence Engineers' Handbook, UK [2] were observed to be inapplicable with Indian geomining conditions which made it necessary to lay down completely separate subsidence formulations for Indian coal mining. Important differences between extractions below uncaved strata and those below caved out goaf(s) were also seen to exist and it was found unsuitable to use the principle of superposition, in its present form, for subsidence estimation in multi-seam workings. The terms single seam and multi-seam used in this paper imply extraction below unfractured strata and that below goaf(s) extracted earlier, respectively.

An important concept, which the Subsidence Engineers' Handbook does not recognise, is that of the non-effective width, this being the maximum width of extraction up to which no significant symptom of subsidence occurs at the surface. This concept is perhaps more important for harder coal measures rather than for softer ones as in the UK. The concept of non-effective width was put forward in India as well as in Australia [3,4].

All the basic concepts, which are well known in subsidence engineering, including the non-effective and critical widths, angle of draw etc, are illustrated in Fig. 1, all extraction widths being expressed as ratios of depth.

The map of India in Fig. 2, shows the locations of the subsidence studies carried out.

2. Discontinuous subsidence

The term discontinuous indicates that the subsidence occurs in shear steps and the subsidence trough or profile cannot be represented by a continuous function. Pot-holing or chimneying is also categorised as discontinuous subsidence but this is not considered in this paper.

The surface damage due to this type of subsidence is particularly severe and as such it was found necessary to specify a limit which could define its occurrence. Fig. 3 shows a line of demarcation between continuous and discontinuous subsidence cases. This line has the equation (depth H and working height h in meters and

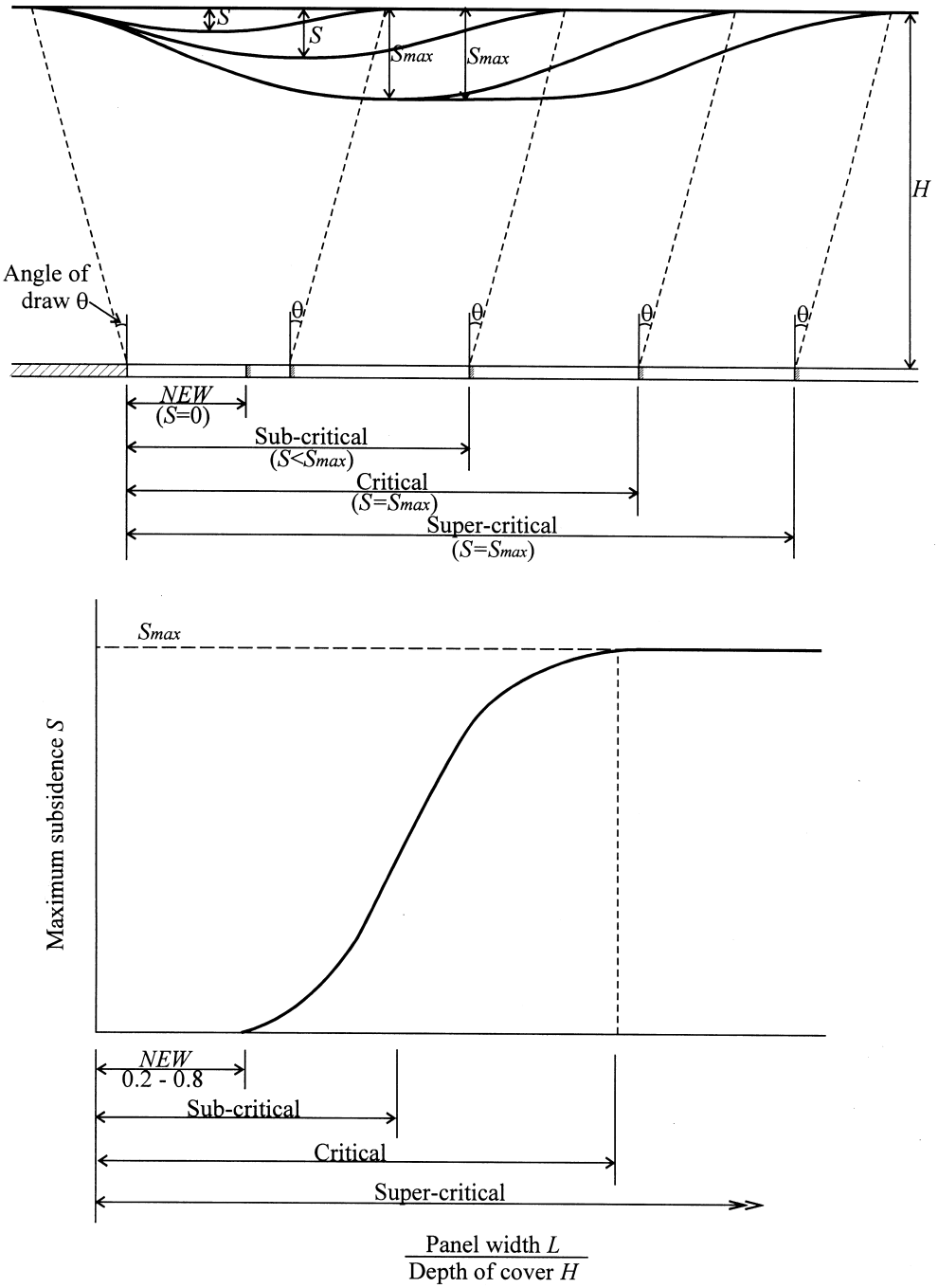


Fig. 1. Schematic for progressive widening of the subsidence trough with panel width defining the basic terms in subsidence engineering.

extraction e in percent)

$$H/h \cdot e = 0.3$$

which covers both single and multi-seam mining conditions. The condition for discontinuous subsidence to occur is thus

$$H/h \cdot e < 0.3 \quad (1)$$

3. Maximum subsidence

The non-effective width-to-depth ratio (NEW) is found to vary approximately between 0.2 and 0.8 in Indian coal fields depending on strata competence (see Fig. 1). Several attempts were made to correlate NEW with strata characteristics but in the absence of complete geomechanical data for the entire strata sections, all such attempts led to poor correlations which were not statistically significant. Consequently, NEW was assigned only approximate values in different parts of the country depending on observation and experience.

In order to plot the maximum subsidence vs the width-to-depth ratio it was necessary to eliminate the variation in NEW so as to lead to a single plot representing different coalfields. Therefore, the maximum subsidence has been plotted vs $x = L/H/NEW$ as shown in Figs. 4 and 5 for single and multi-seam extractions. Only those cases in which the NEW was actually measured have been selected for these two plots. The statistical best fit curves are expected to

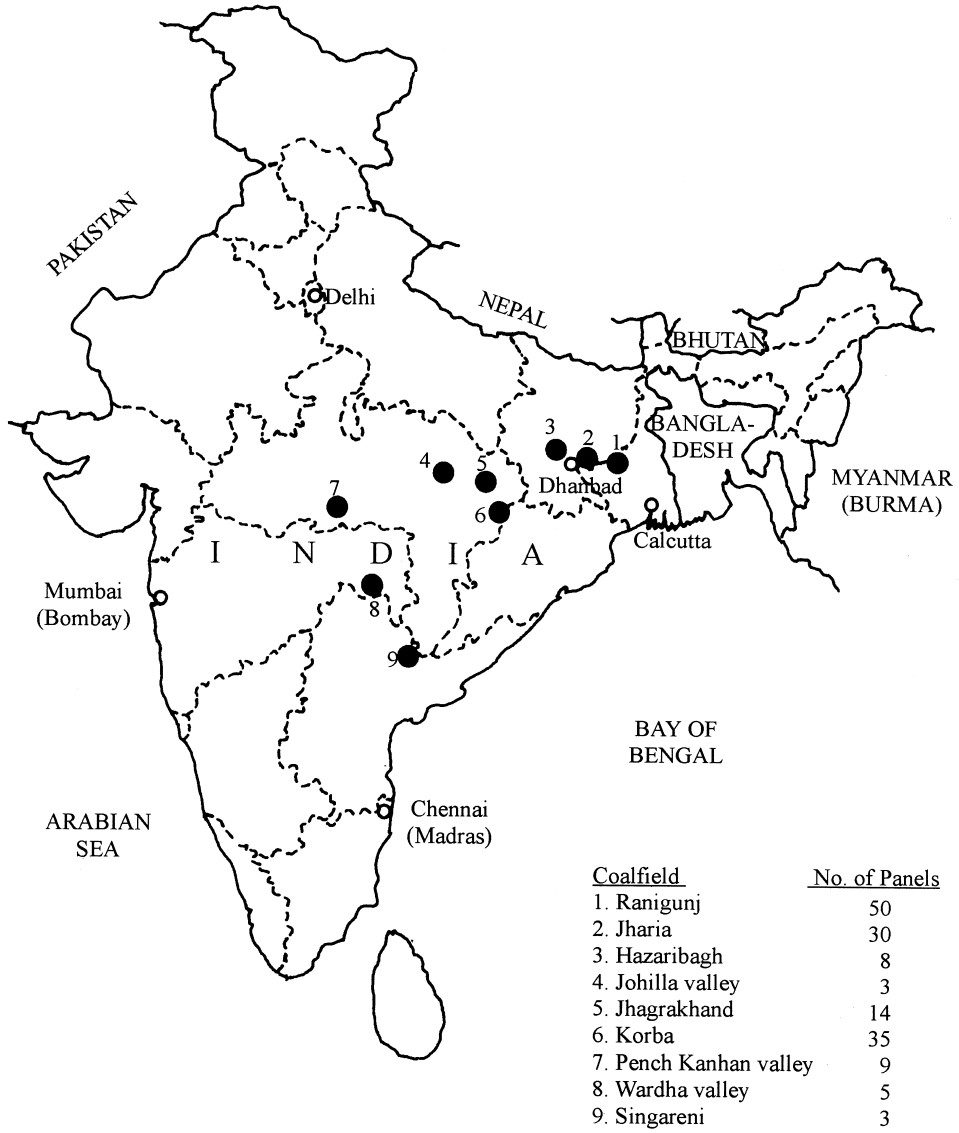


Fig. 2. Map of India showing the coalfields where subsidence studies were conducted.

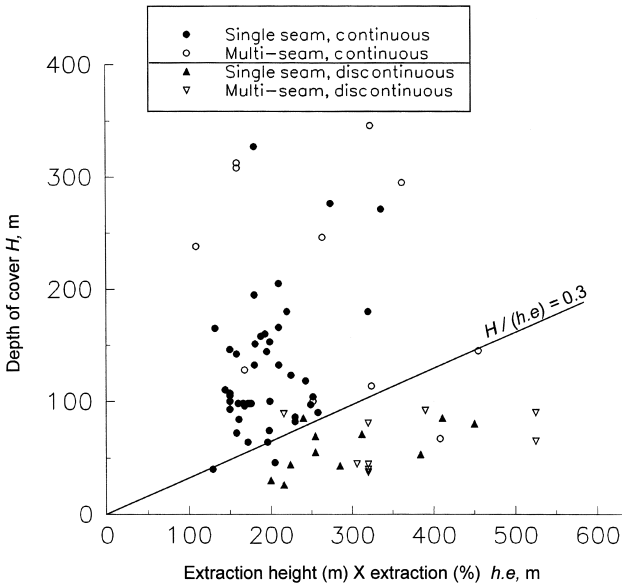


Fig. 3. Line of demarcation between continuous and discontinuous subsidence.

have two inflexion points [2] and must become asymptotic beyond the critical width. A hyperbolic tangent function is logical for such curves. The general equation which would fit these plots has thus been chosen in the form

$$S = c_1 \cdot h \cdot e\left[1 - \frac{\tanh\ c_2(x - c_3)}{\tanh\ c_2(1 - c_3)}\right] \quad (2)$$

where c_1, c_2 and c_3 are constants. During this exercise, it was seen that several cases from central India gave the subsidence factor ($S_{max}/(h{\cdot}e)$) as 0.3–0.35. Such unusually low values have not been reported from anywhere else in the world. In addition, while fitting the influence function model to many single seam cases, higher values of the subsidence factor were really required. These cases of low subsidence factor, therefore, had to be regretfully rejected until such time that the reasons for such low subsidence could be established. It was possible that subsidence observations in these cases were not carried out for long enough. Also eliminated from these plots are the cases of discontinuous subsidence.

It may be seen from Figs. 4 and 5 that the subsidence factor increases from 0.69 for single seam cases to 0.81 for multi-seam cases.

Eqn (2) is not convertible-to-linear for the purpose of regression. A non-linear regression analysis was therefore programmed employing the "search method" of solution which would satisfy the law of least squares. This led to the following two relations

- single seam cases

$$S = 0.33[1 + 1.1\ \tanh(1.4(x - 1.8))] \quad (3)$$

- multi-seam cases

$$S = 0.12[1 + 5.9\ \tanh(0.38(x - 1.45))] \quad (4)$$

The index of determination (R^2) was obtained as 0.85 and 0.86, respectively. It is obvious that eqns (3) and (4) are significantly different. A word of caution regarding eqn (4) is necessary. It may be noted that in multi-seam room and pillar extractions the overlying

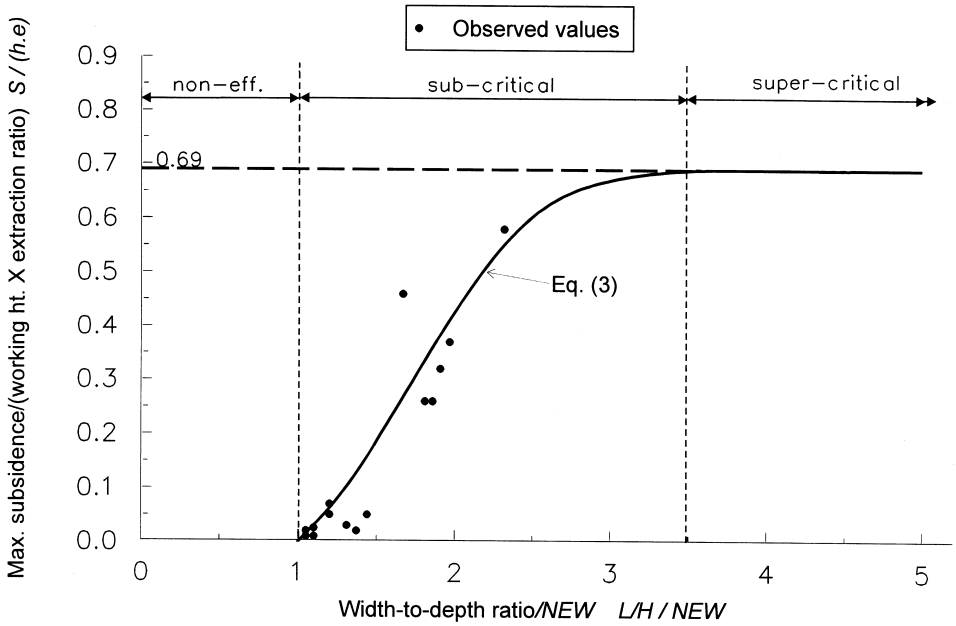


Fig. 4. Maximum subsidence for single seam cases.

goaf(s) may or may not overlap the extraction panel and also can have widely different and irregular shapes. Eqn (4), therefore, is only approximate and it will always be prudent to use the influence function method for assessing the complete subsidence in a given multi-seam situation.

The critical width of extraction is found to be $3.5 \times NEW$ in single seam cases while in multi-seam cases it is found to be 6.5–7 times the *NEW*. This clearly happens because of a fall in the *NEW* value in multi-seam cases.

When seams are extracted one after the other in descending order, the principle of superposition for obtaining the total surface subsidence can be used keeping in mind that eqn (3) is applicable only to the top seam, which is extracted below unfractured rock, while eqn (4) has to be used for all lower seams.

It was felt necessary to compare subsidence over hydraulically stowed panels with that over caved panels. Fig. 6 shows those single seam cases where stowing was done well and where *NEW* was also measured. There were several instances, however, where the subsidence observed was much greater than as depicted in Fig. 6. These were clearly cases of poor stowing. Since subsidence prediction becomes impossible if stowing is badly done, it is possible to give an equation only for good stowing. However, seeing the scatter in Fig. 6 no equation is given. As an approximation, a subsidence factor of 0.05 is proposed here for good stowing.

4. Angle of draw

In subsidence engineering, the angle of draw (see Fig. 1) is expressed as a geomining property and should therefore be obtained free of any bias due to mining geometry. Ideally, true draw is estimated from a long rectangular panel, the subsidence profile being plotted across the smaller width. This automatically means that true draw must be measured normal to the panel side (in plan).

Three examples of influence of the panel shape in plan can be seen from the 5-mm contour, which defines the limit of the observed subsidence trough, in Fig. 7. The following points about draw can be readily seen from this figure:

- the draw limit is significantly affected around corners;
- over panels only slightly wider than *NEW* the draw is negative at the shorter sides and positive at the longer sides;
- an extraction panel must have a sufficient width and length for the draw to develop all around the panel.

The draw angle varies considerably in India with strata conditions (9–25°) and, like *NEW*, it was not possible to correlate draw with any geomechanical index. Simi-

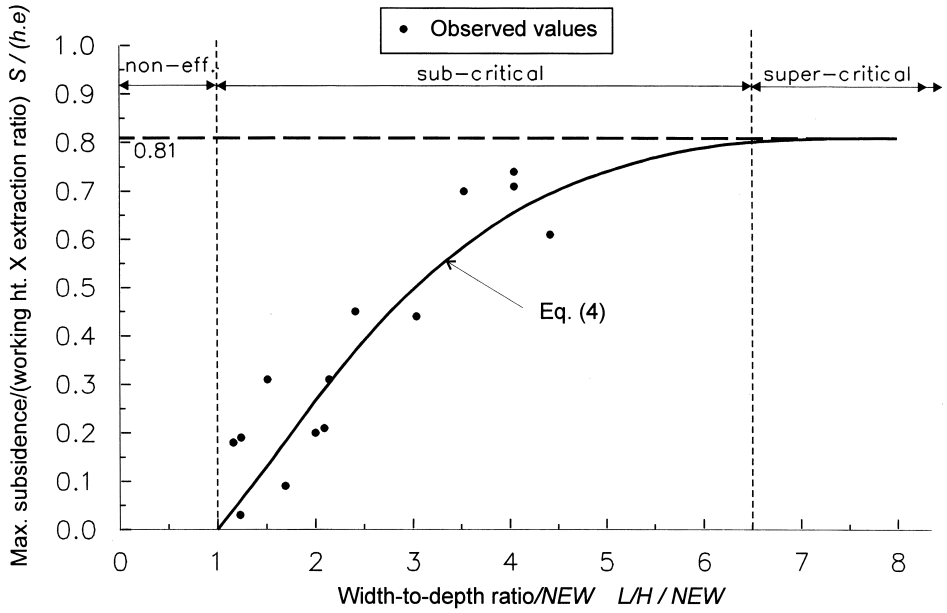


Fig. 5. Maximum subsidence for multi-seam cases.

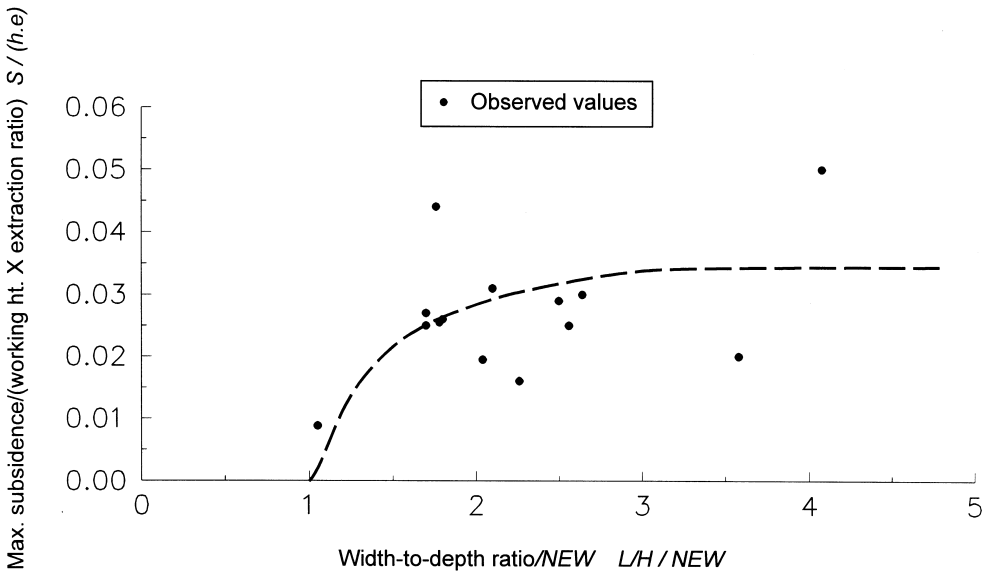


Fig. 6. Maximum subsidence for stowing cases.

larly, observations over cases with steep dips were few and far between and a satisfactory relationship between angle of draw and seam dip could not be derived.

5. Maximum slope and strains

Standard relationships for maximum slope and strains in relation to the maximum subsidence are given in the Subsidence Engineers' Handbook which are as follows.

$$G = K_1 \frac{S}{H}$$

$$E_{(-)} = K_2 \frac{S}{H}$$

$$E_{(+)} = K_3 \frac{S}{H}$$

From these relations the constants K_1, K_2 and K_3 can be calculated from subsidence, slope and strain measurements at the surface. Figs. 8 and 9 are once again plotted with the variable $L/H/NEW$ as the abscissa. The two plots are given separately for single and multi-seam cases. It can be noticed that the number of cases in Fig. 8 is considerably larger than in Fig. 4 because Fig. 8 includes cases of hydraulic stowing as well as those cases from central India where abnormally low subsidence factors were observed.

In Figs. 8 and 9 the curves are hand drawn and have not been obtained as best fits from regression analysis.

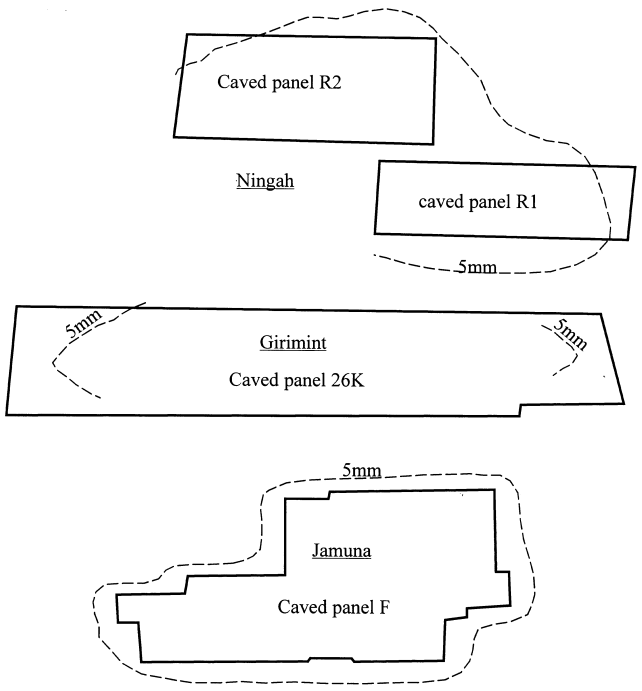


Fig. 7. Influence of panel plan geometry on draw limit as shown for three mines.

6. A modified influence function method

Much work has been done for modelling subsidence employing the influence function method and different functions have been proposed in the past. Several relatively recent publications [5–9] have dealt with the influence function method.

6.1. Conventional method

The method basically consists of laying a grid on the surface covering the extraction panel area and extending up to the subsidence draw limits. An influence area of circular shape at each grid point is divided into a sufficient number of rings, each ring being subdivided into a sufficient number of equal sectors by drawing radii from the centre of the influence area which is also the grid point (Fig. 10). The radius of the influence circle is to be taken as $H \tan \theta$. The sector which falls inside the excavation will give rise to some subsidence at the grid point P, depending on the weighting factor of that sector, and the sector which falls outside the excavation gives no subsidence at P.

The weighting factor for the ith ring is given by

$$S_i = \iint_A K_z r \, d\alpha \, dr \tag{5}$$

where A represents the area of the annulus or ring. Ren et al. [5] have recommended 10 rings of equal width and 64 sectors or elements in each ring giving 640 elements in the influence circle. It this is adopted the weighting factor for each element will be obtained as $S_{ie} = S_i/64$.

The subsidence at point P is simply the sum of all the element weighting factors which fall in the extraction panel multiplied by the maximum possible subsidence S_{max}.

The weighting factor for the elements falling outside the extracted panel is zero.

The horizontal displacement at point P due to an extracted element is

$$dV_{xy} = \tan \zeta dV_z \quad (dV_z = S_{ie} S_{max})$$

where dV_z is the vertical movement at P due to the extracted element and ζ is the angle between the vertical and the line joining the element centroid with point P (Fig. 10). The total horizontal movement is again the sum of the movements due to all extracted elements.

The components of these movements along x and y are obtained as

$$dV_x = dV_{xy} \cos \beta$$

$$dV_y = dV_{xy} \sin \beta$$

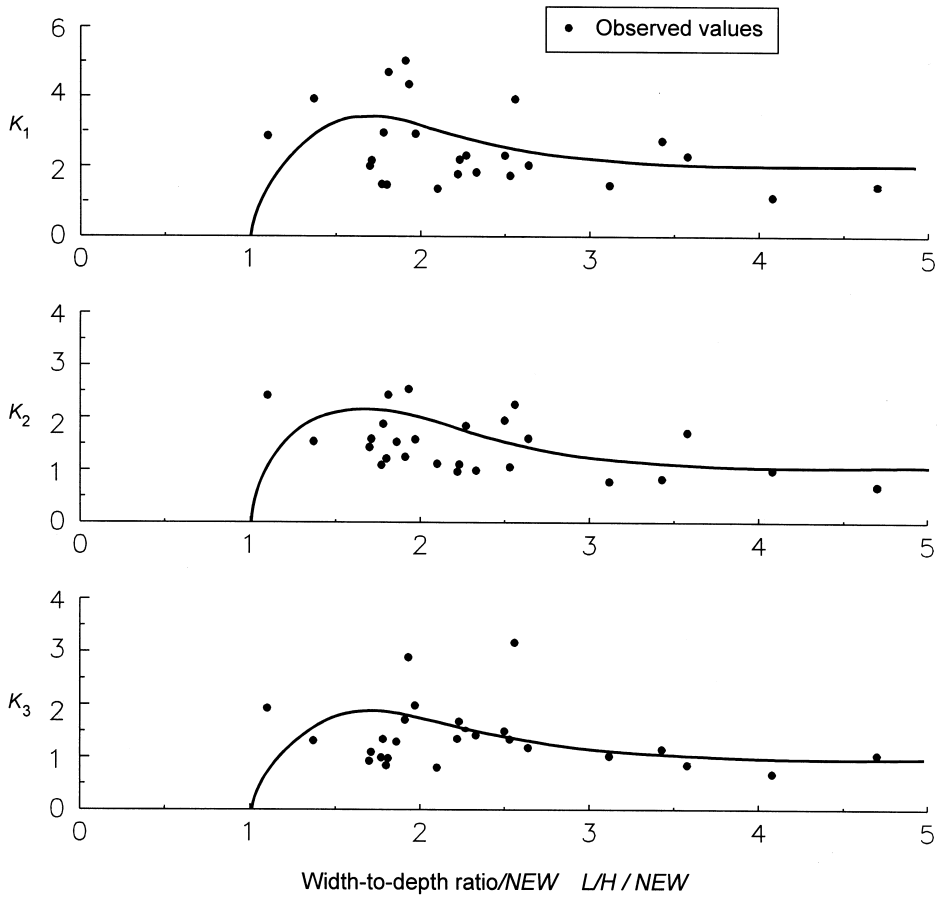


Fig. 8. Reference graph for estimation of K_1, K_2 and K_3 for single seam cases.

where β is the angle shown in Fig. 10. The total displacement of P is again a sum of those due to all extracted elements.

The slope and strain at point P are then readily calculated having known the vertical and horizontal movements.

6.2. Modification of the method

The following influence function

$$K_z = \frac{0.5352}{R^2}\left(1 + \cos\frac{\pi \cdot r}{R}\right) \tag{6}$$

was found to be suitable while simulating surface subsidence. An influence function can have various forms, since the method is really empirical and the choice of a function is arbitrary, but it must satisfy the condition

$$\iint_{A_0} K_z \, dA = 1 \tag{7}$$

where A_0 is the area of the influence circle.

Substituting eqn (6) in eqn (5), the ring weighting factors were obtained as given below.

i	1	2	3	4	5	6	7	8	9	10
S_i	0.0297	0.0854	0.1311	0.1607	0.1708	0.1596	0.1294	0.0859	0.0397	0.0069

The sum of the above weighting factors is 1.0 and thus eqn (7) is satisfied.

The conventional method as described earlier gives rise to much higher subsidence at the edges of a panel, as reported earlier [5,6]. This was also experienced by the authors of this paper. Ren et al. [5]

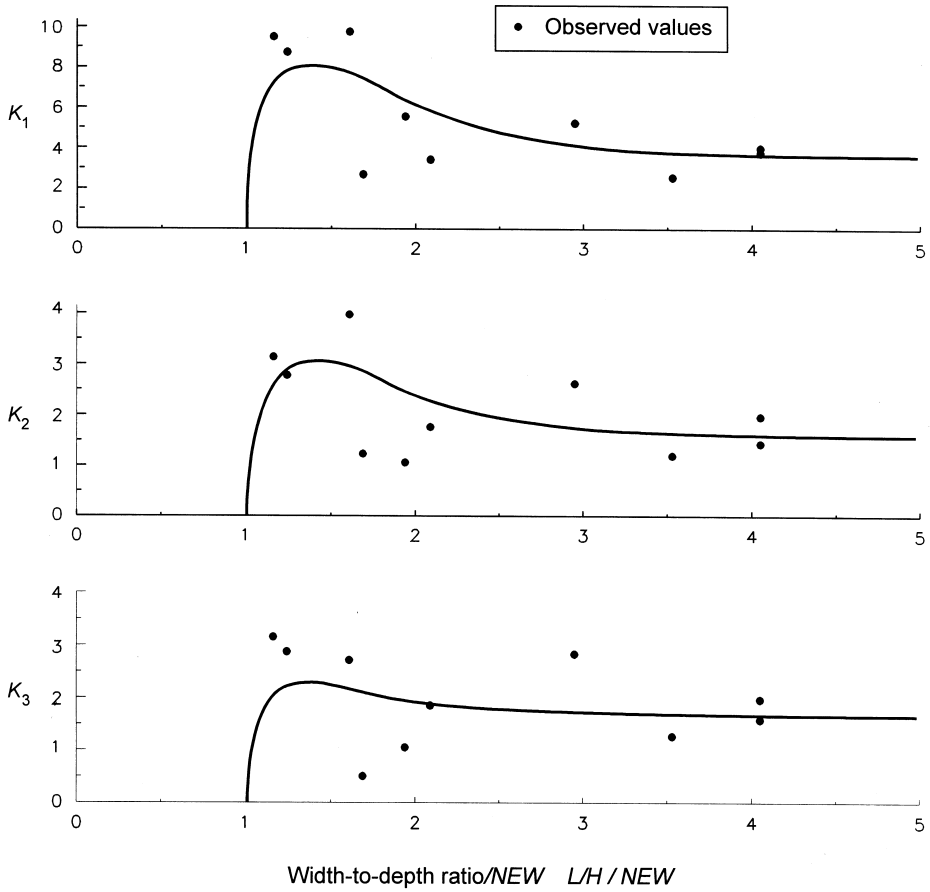


Fig. 9. Reference graph for estimation of K_1, K_2 and K_3 for multi-seam cases.

have suggested drawing an artificial boundary some distance inside the panel on all sides and treating this new boundary as the extraction panel. Liao [6] has suggested fuzzifying the influence function closer to the panel edges.

Besides the correction to the influence function for panel edges, another correction required was to account for the asymmetry of the subsidence trough which is invariably found in Indian conditions. The point of maximum subsidence always occurs closer to the start line of the panel.

The correction function (weighting function) has been suitably designed as

$$W_z = 0.5 \ \tanh\left(\frac{5d_b}{1.5NEWh} - 2.4\right) + 0.5 \tag{8}$$

where d_b is the distance of the extraction element to the nearest boundary of the extracted area.

The correction function for asymmetry has been designed as follows

$$Q_z = 0.9 - 0.1 \ \tanh[0.5(d - 0.4d_{max})] \tag{9}$$

where d is the distance of the extraction element centroid from the starting face line and d_{max} is the distance between starting and ending face line.

We thus have the modified influence function using eqns (8) and (9) as

$$dV'_z = Q_z W_z dV_z$$

6.3. Influence of overlying goaf

As mentioned earlier, overlying goaf(s) influence the surface subsidence, changing the subsidence factor as well as the maximum slope and strains. After some hit and miss trials it was found that a 40% reduction in the original *NEW* value along with a 17% increase in the subsidence factor enables prediction of multi-seam subsidence to a reasonable accuracy. Since the overlying goaf may or may not completely overlap the extraction panel, these changes in *NEW* and subsidence factor would be applicable only for those elements falling within the goaf area. Moreover, since the parting between the extraction panel and the overlying goaf is intact, the final evaluation is done by taking a weighted average as

$$NEW' = \frac{0.6NEW(H - T) + NEW(T)}{H}$$

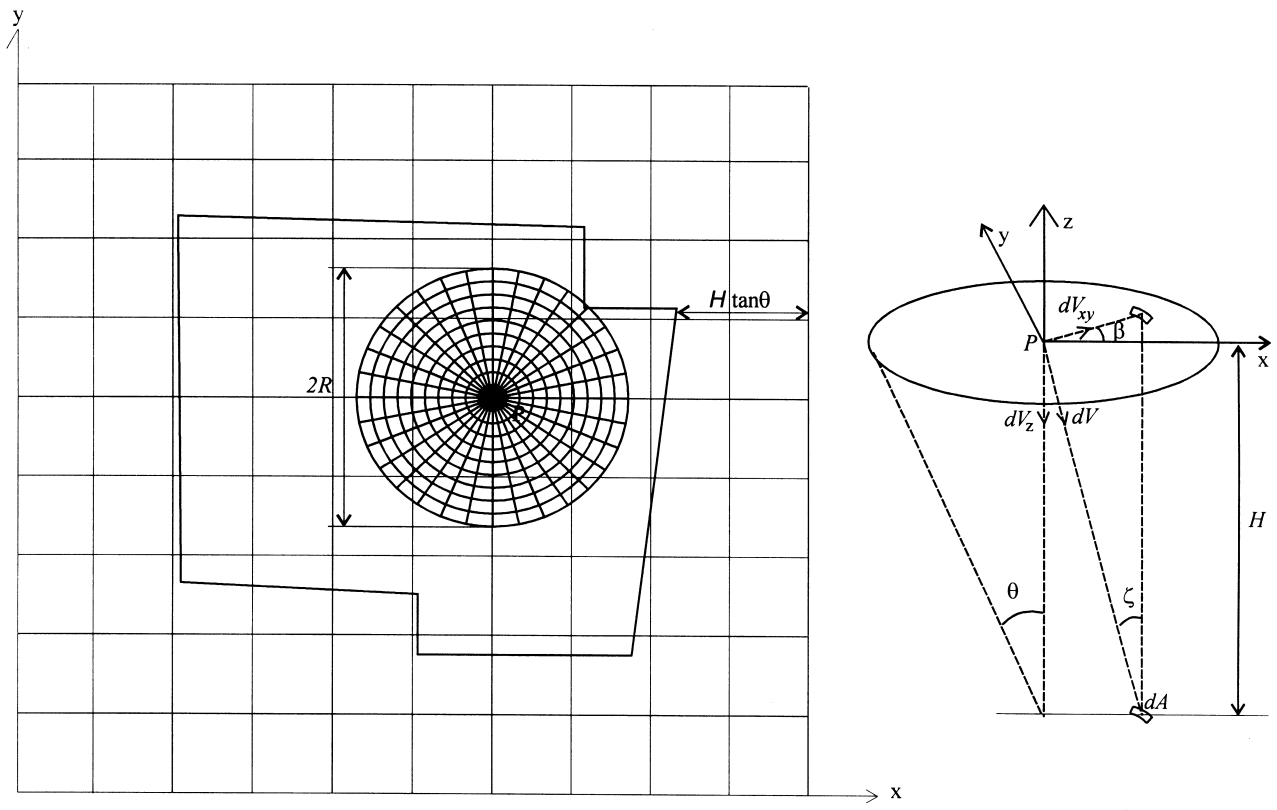


Fig. 10. Illustration of subsidence and horizontal displacements calculation using influence function method.

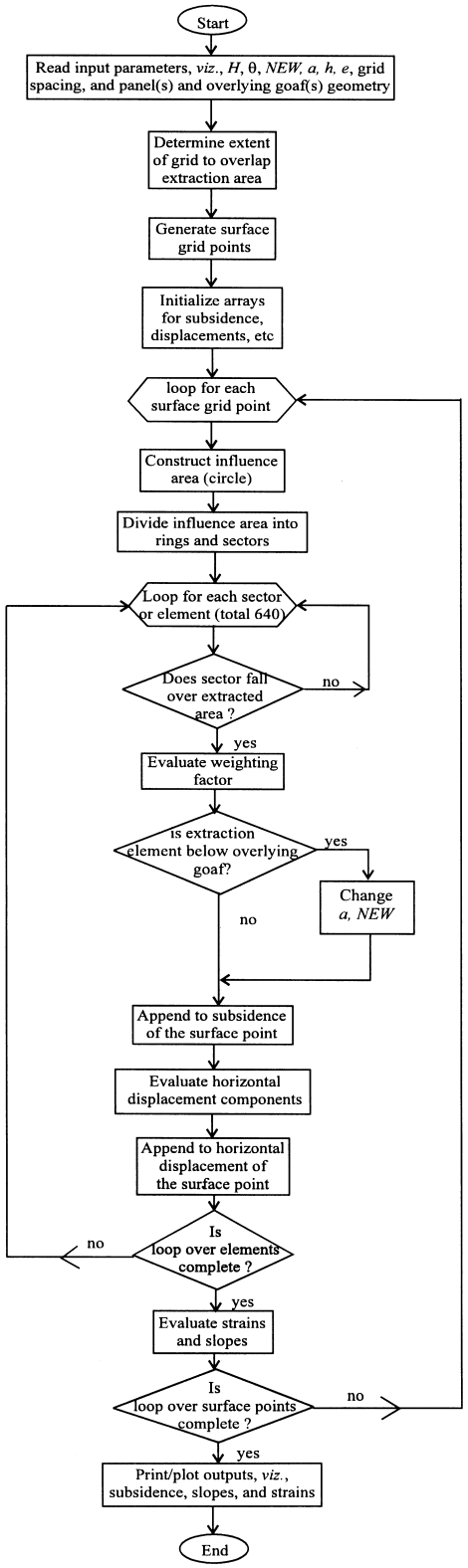


Fig. 11. Flow chart representation of computer program.

$$a' = \frac{1.17a(H-T)+a(T)}{H} \tag{10}$$

where T is the parting between the extraction and overlying goaf.

It has been mentioned earlier that the average subsidence factor increases from 0.69 for single seam cases to 0.81 for multi-seam cases, i.e. by about 17%. This increase represents observations over various rock parting thicknesses and also various goaf shapes and sizes. The second part of eqn (10) assigns a separate weightage to the intact rock parting contrary to the above finding. This equation, however, was adopted after several trials of applying the influence function method to various cases.

7. Comparison between predictions and actual observations

7.1. Subsidence contours

A computer program has been developed for this modified influence function method both for single and multi-seam extraction. A flow chart of the computer program is given in Fig. 11. To check the validity of the method, eight single seam cases and five multi-seam cases were taken. The comparison between the predicted and observed subsidence values was found to be satisfactory. Only two single seam cases and two multi-seam cases are shown as examples in this paper. In room and pillar cases, pillar remnants which are left underground can sometimes give some difference between the predicted and observed values. Unfortunately, information regarding such left out remnants, if any, could not be obtained in any of the cases from mine records.

For sufficient clarity, the predicted and observed values have been compared as contours on panel plans and as subsidence profiles in the x and y directions.

7.2. Single seam cases

The two single seam cases presented include one room and pillar extraction panel and two adjacent longwall panels. The subsidence factor for all the single seam cases is taken as 0.69, as concluded from Fig. 4.

7.2.1. Sukri colliery

The panel shown in Fig. 12 was a room and pillar extraction panel at an average depth of 165 m. The moderately dipping seam (1 in 5.2) was worked out to a height of 2.4 m. An extraction of 75% is taken for

calculations as supplied by the colliery management. From field observations, *NEW* and angle of draw were found to be 0.45 and 13°, respectively. Fig. 12(a) shows the predicted surface subsidence contours for this panel. For comparison the subsidence contours observed from field measurements are presented in Fig. 12(b). For influence function modelling, highly irregular boundaries (refer to Fig. 12(b)) are reasonably approximated with straight lines as shown in Fig. 12(a). Fig. 12(c) shows a comparison of two subsidence profiles.

7.2.2. Ningah colliery

At Ningah colliery, two longwall panels were worked out at an average depth of 180 m as shown in Fig. 13. The height of extraction for these panels was 2.0 m, the extraction being 100%. Field observations gave a non-effective width-to-depth ratio of 0.7 and an angle of draw of 20°. These two panels had widths very close to the non-effective width observed in this area. The predicted and observed subsidence contours are presented in Fig. 13(a) and (b) respectively and the subsidence profiles in Fig. 13(c).

7.3. Multi-seam cases

In both the multi-seam cases presented in this paper the overlying goaf(s) partly overlap the working panel(s). The subsidence factor has been evaluated specifically for each grid point depending on its position with respect to the overlying goaf and the parting

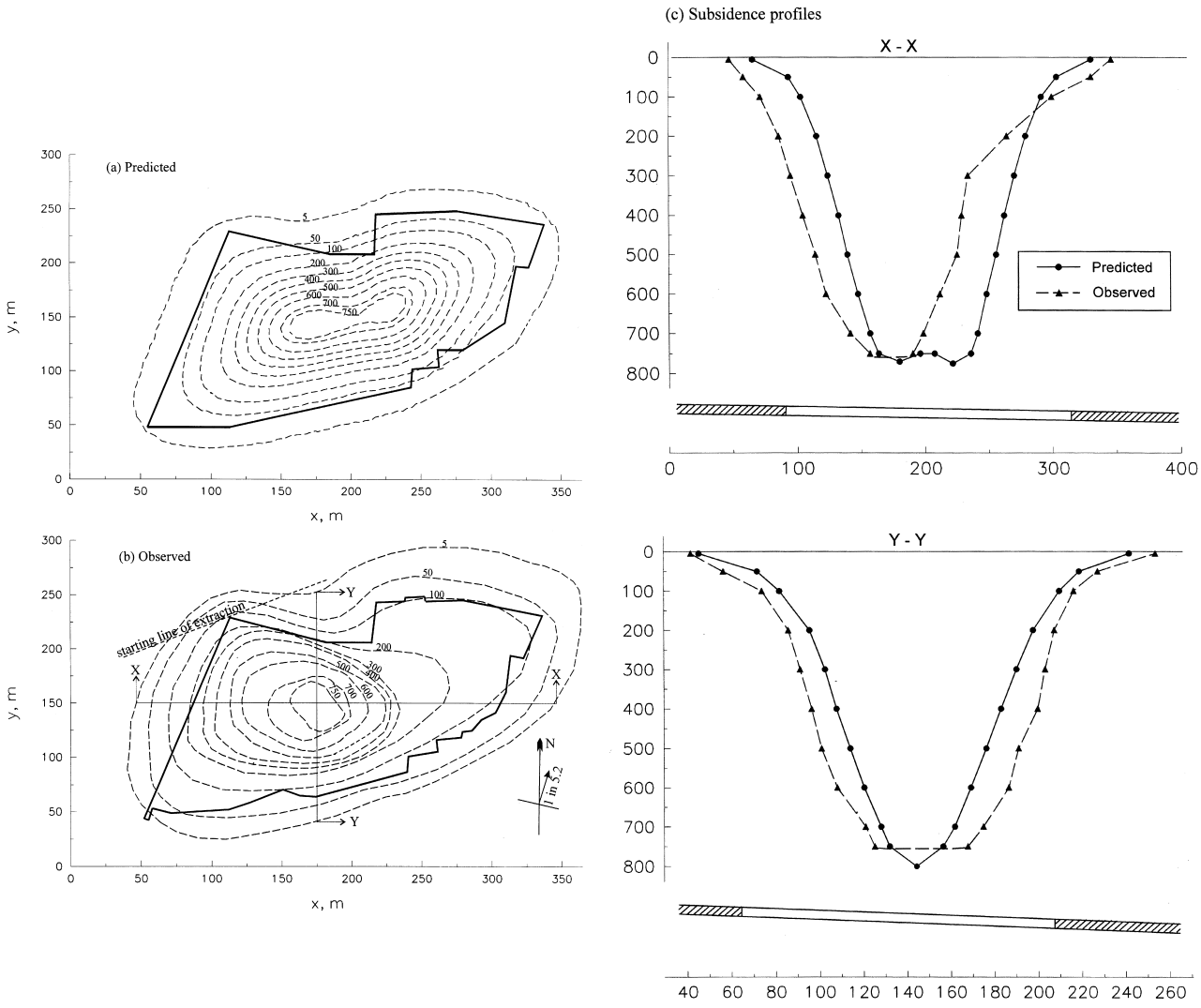


Fig. 12. Comparison of subsidence at Sukri colliery: (a) predicted; (b) observed contours (mm) over the panel plan; and (c) subsidence profiles at sections xx and yy shown in (b).

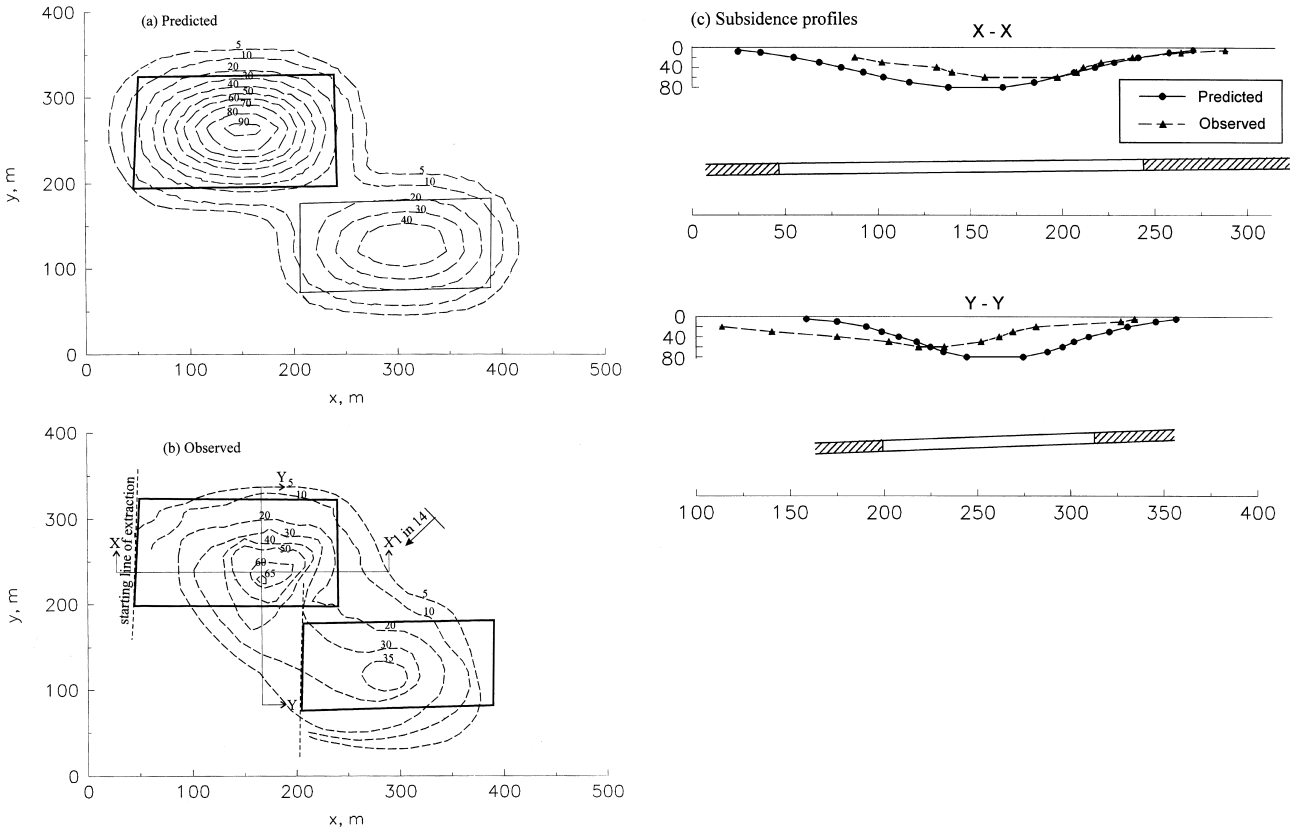


Fig. 13. Comparison of subsidence at Ningah colliery: (a) predicted; (b) observed contours (mm) over the panel plan; and (c) subsidence profiles at sections xx and yy shown in (b).

between the working and the goaf. In both cases, the highly irregular boundaries of working, as well as overlying goaf, are approximated using straight lines for simplifying the computer modelling.

7.3.1. Bera colliery

At this mine, the room and pillar extraction panel in seam II shown in Fig. 14 was at an average depth of 67 m. Seam II was worked to a height of 4.8 m with an extraction of 85%. The overlying goaf of seam III partially overlaps the seam II workings as shown in Fig. 14 and occurs at a parting of 19 m. The *NEW* for intact strata was observed as 0.32 and the angle of draw was found to be 20°. The predicted and observed subsidence contours for this case are presented in Fig. 14(a) and (b) respectively and the subsidence profiles in Fig. 14(c).

7.3.2. North Searsole colliery

Two room and pillar panels in Kenda bottom seam here were extracted side by side as shown in Fig. 15. The workings are at an average depth of 45 m from the surface and Kenda top seam is above Kenda bottom seam at a parting of 19 m. The boundary of overlying goafs in Kenda top seam is shown in Fig. 15. Kenda bottom seam was worked out to a height of 4 m with an extraction of 75%. The observed angle of draw is 20° and *NEW* is 0.32. Fig. 15(a) and (b) show the predicted and observed subsidence contours, respectively, for this case and Fig. 15(c) depicts a comparison of subsidence profiles. Along section Y–Y there is a clear indication of a remnant pillar which has not been reported.

7.4. Slope and strain

One case each for single and multi-seam conditions illustrates the comparison between observed slope and strain variation as shown in Figs. 16 and 17. The names of the mines and other relevant data are also given in the same figures.

8. Discussion and conclusion

The influence function method with suitable modifications, as mentioned in this paper, for subsidence asymmetry, edge effect and overlying goaf(s) is found

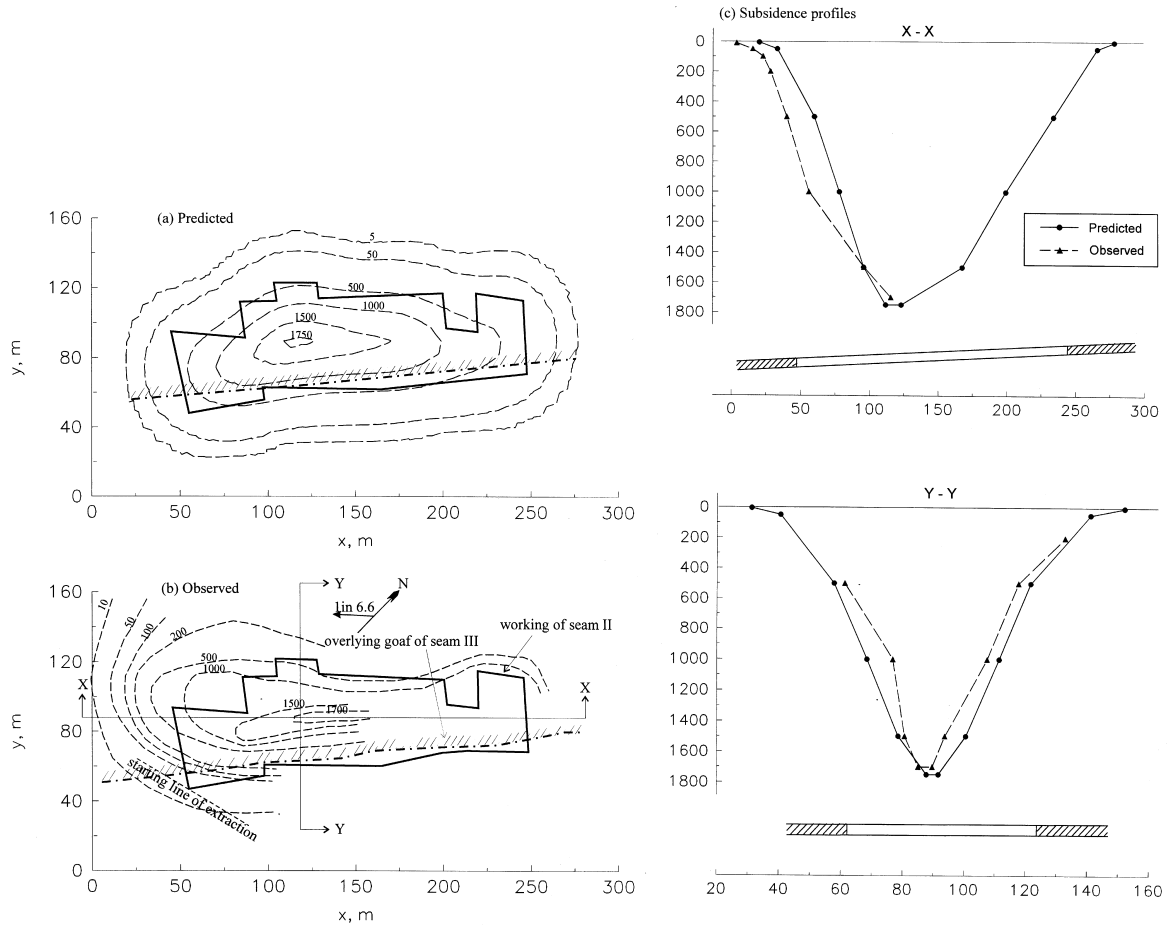


Fig. 14. Comparison of subsidence at Bera colliery: (a) predicted; (b) observed contours (mm) over the panel plan; and (c) subsidence profiles at sections xx and yy shown in (b).

to be reasonable for all panel shapes and sizes. While trying to fit this model to the observed cases, it was found that the effect of corners on draw was not completely satisfactorily represented. Secondly, when a panel had a non-effective width some small subsidence was obtained, although the difference was rather small. Thirdly, the occurrence of negative draw along the shorter sides of a long narrow panel, whose width is slightly more than the non-effective width, could not be simulated by the influence function method.

The influence function method as described in this paper is for horizontal or near horizontal seams and when the gradient starts influencing subsidence, the predictions do not agree very well with the observations. As an example, this can be seen from the case of Sukri colliery (Fig. 12(b)) where the gradient is 1 in 5.2. The authors intend to further the work and develop a suitable model for inclined seams.

The line demarcating the limit of discontinuous subsidence has been given for single seam as well as multi-seam cases, but it is felt that the thickness of unfractured rock in relation to the total depth in multi-seam cases will probably make some difference to this limit.

A definite increase in the subsidence factor and also slope and strains is noticeable while extraction below goaf(s) is carried out.

The occurrence of unusually low subsidence in some of the central Indian coal mines cannot be explained properly, although it is conjectured that the observations were not carried out for long enough. Such low subsidence may be related to several causative factors. It is possible that the in situ stress field may be related to the magnitude of subsidence, although such studies have not been done in the past. Another factor that may be responsible is the bulking factor upon caving. Some studies may be done in the future to identify the factors responsible for such low subsidence.

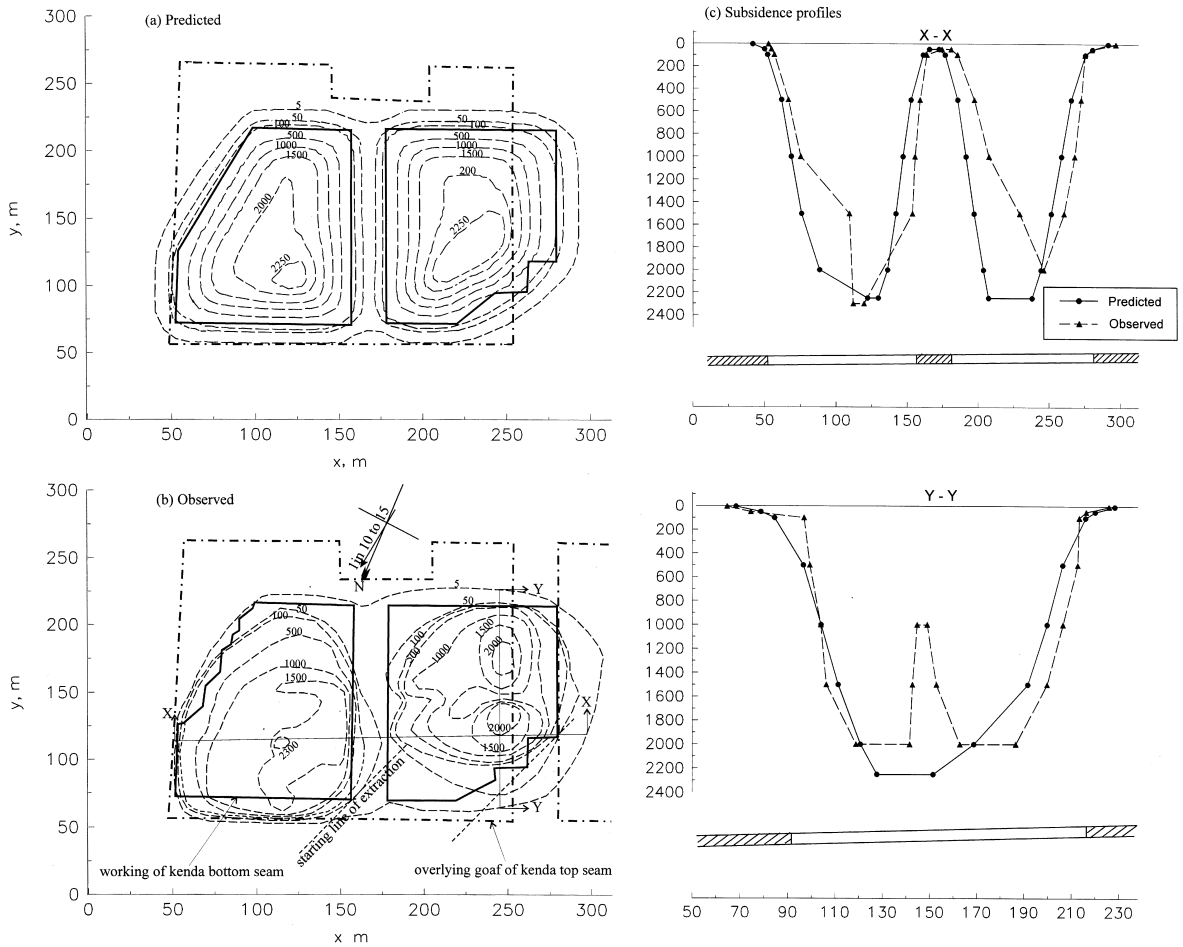


Fig. 15. Comparison of subsidence at North Searsole colliery: (a) predicted; (b) observed contours (mm) over the panel plan; and (c) subsidence profiles at sections xx and yy shown in (b).

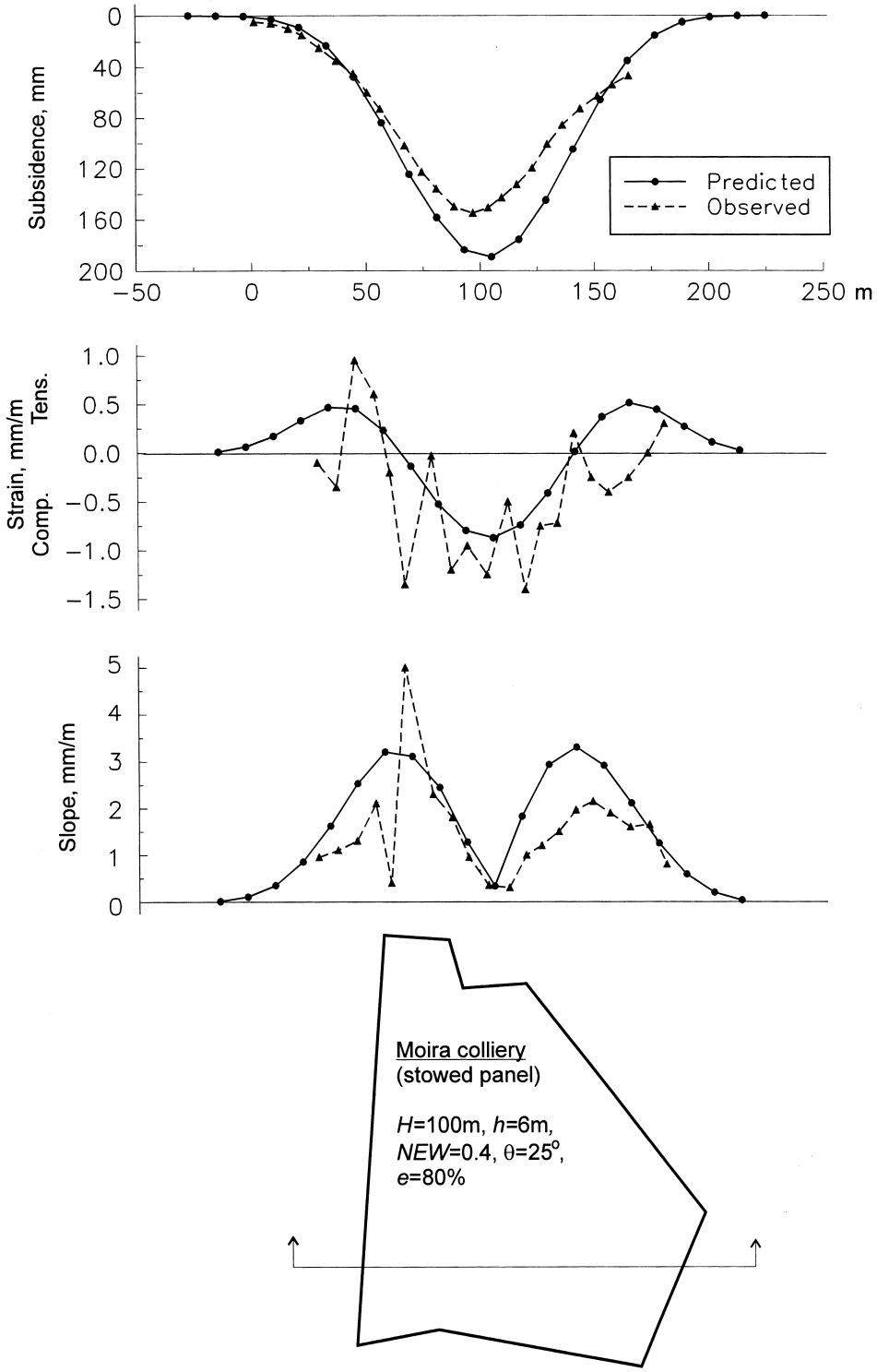


Fig. 16. Subsidence, strain and slope profiles at Moira colliery (panel plan shown at the bottom by solid lines).

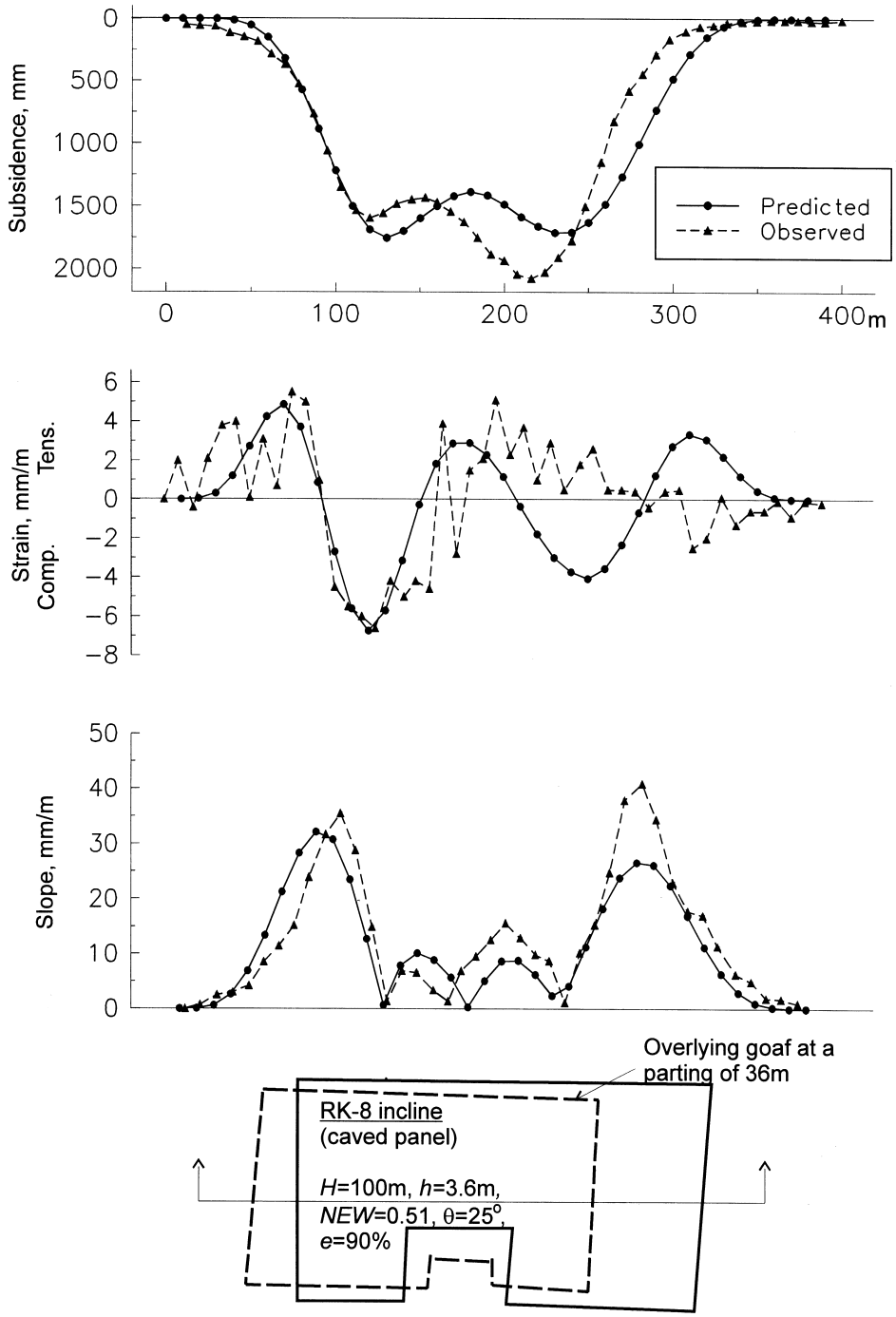


Fig. 17. Subsidence, strain and slope profiles at RK-8 incline (panel plan shown at the bottom by solid lines).

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